

TREE-01

Q9)

Question:

DS	Session	Tree 1	Question Information	Level 1	Challenge 69
Question description A forest is an undirected graph without cycles (not necessarily connected). Mohana and john are friends in kerala, both of them have a forest with nodes numbered from 1 to n, and they would like to add edges to their forests such that: • After adding edges, both of their graphs are still forests. • They add the same edges. That is, if an edge (u,v) is added to Mohana's forest, then an edge (u,v) is added to john's forest, and vice versa. Mohana and johns want to know the maximum number of edges they can add, and which edges to add. Constraints: $1 \leq n \leq 105,$ $0 \leq m \leq 1$ $m \leq n$ $1 \leq u, v \leq n, u \neq v$					

PROGRAM:

```
1  #include <stdio.h>
2
3  int p[100005],q[100005];
4
5  int find(int a[],int x){
6      if(a[x]!=x) a[x]=find(a,a[x]);
7      return a[x];
8  }
9
10 int main(){
11     int n,m1,m2,u,v,i,j;
12
13     scanf("%d%d%d",&n,&m1,&m2);
14
15     for(i=1;i<=n;i++){
16         p[i]=q[i]=i;
17     }
18     for(i=0;i<m1;i++){
19         scanf("%d%d",&u,&v);
20         p[find(p,u)]=find(p,v);
21     }
22
23     for(i=0;i<m2;i++){
24         scanf("%d%d",&u,&v);
25         q[find(q,u)]=find(q,v);
26     }
27
28     printf("%d\n",n-(m1>m2?m1:m2)-1);
29
30     for(i=1;i<=n;i++){
31         for(j=i+1;j<=n;j++){
32             if(find(p,i)!=find(p,j) && find(q,i)!=find(q,j)){
33                 printf("%d %d\n",i,j);
34                 p[find(p,i)]=find(p,j);
35                 q[find(q,i)]=find(q,j);
36             }
37         }
38     }
39
40     return 0;
41 }
```

OUTPUT:

TEST CASE 1:

```
5 3 2
5 4
2 1
4 3
4 3
1 4
1
1 5
```

TEST CASE 2:

```
5 3 2
5 3
2 3
2 3
2 3
2 4
1
1 2
4 5
```